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Extension Tectonics: The Neogene Opening of the North-South Trending Basins of Central Thailand

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Paleomagnetic samples were collected from late Neogene basalt flows from Thailand. All of these flows are horizontal and are relatively unaltered in thin section. These rocks possess a stable magnetization which is believed to be primary. Samples from 48 lava flows were collected from sites located within the Khorat Plateau, the Chao Phraya-Phitsanulok Basin, and the mountainous terrane west of the Chao Phraya-Phitsanulok Basin. These data were combined with previously reported late Neogene data from five flows from western Thailand. Although the average inclination from the 53 sites is indistiguishable from the expected dipole inclination, the average declination has a net clockwise rotation of $13.5^\circ \pm 5.8^\circ$ from the geocentric dipole field. Furthermore, the mean declination values from the 29 flows from the Khorat Plateau are indistinguishable from the present dipole field direction ($D_m = 4.3^\circ \pm 7.5^\circ$) and indistinguishable from the mean declination from 28 late Neogene volcanic flows from Vietnam. In contrast, the mean declinations from 24 flows collected from central and western Thailand are deflected significantly clockwise ($D_m = 24.4^\circ \pm 7.7^\circ$) from the geocentric dipole field direction. The differential rotation between western and central Thailand versus the Khorat Plateau suggests that Indochina is composed of at least two structural blocks which underwent a different rotational history. These observations, when combined with geologic and geophysical data from the Chao Phraya-Phitsanulok Basin, Gulf of Thailand, and the intermontane basins of western Thailand, suggest that the rotations are recording a late Neogene phase of E-W extension of these basins. We suggest that the formation of these basins and the related basaltic volcanism developed in reponse to subduction of the Indian plate under western Burma. We envision the tectonics of this region is similar in style to the Basin and Range region of the western United States. Last, we have observed field relationships from some of the rhyolites located in the central basin. Although these rhyolites are reported to be Mesozoic or Paleozoic in age, our field observations and a K-Ar age date show that at least some of these rhyolites are younger than the basalts. We suggest that the rhyolites form a bimodal suite with the basaltic rocks which were erupted in the later stages of the extension.

INTRODUCTION

Numerous fault-bounded basins trend north-south in an en echelon fashion throughout central Thailand. These basins, which become progressively wider toward the south, are (1) the intermontane basins in northwestern Thailand, (2) the Chao Phraya-Phitsanulok Basin in central Thailand, and (3) the Gulf of Thailand (Figure 1). All of these basins are filled with thick sequences of Tertiary and Quarternary siliciclastic rocks which reach a maximum thickness of approximately 8 km in the Gulf of Thailand. Various models for the time of formation of these basins have been proposed. On the basis of the thickness of the Tertiary sections in the intermontane basins and in the Gulf of Thailand, *Achalabhuti* [1976] proposed that these basins began forming in the early Tertiary and that subsidence has been continuous thoughout the entire Tertiary. *Suensilpong et al.* [1978] propose an even earlier age of formation, suggesting the western intermontane basins formed by uplift during the Cretaceous and early Tertiary. *Hamilton* [1979] and *Hellinger et al.* [1981] also propose an early Tertiary initiation of basin formation in the Gulf of Thailand. *Tapponnier et al.* [1982] suggested a model for the formation of the Gulf of Thailand in which the opening occurs post-early Tertiary, after the Eocene collision between India and Eurasia.

The models proposing an early Tertiary formation for these basins assume that the oldest sediments contained in these basins are of early Tertiary age. However, most of the sediments are nonmarine and lack diagnostic age fossils. In many basins (e.g., the northern Mae Tip and Li basins), the ages are based on stratigraphic correlation with other poorly dated pre-Neogene sections. The lack of well-dated fossil horizons makes us question the reliability of the early Tertiary age of basin formation.

The following geologic and geophysical data from the region suggest that the basins opened much later: (1) north-south trending normal faults with approximately 100 m of throw cut Pleistocene rocks in the Gulf of Thailand [*Tra-*

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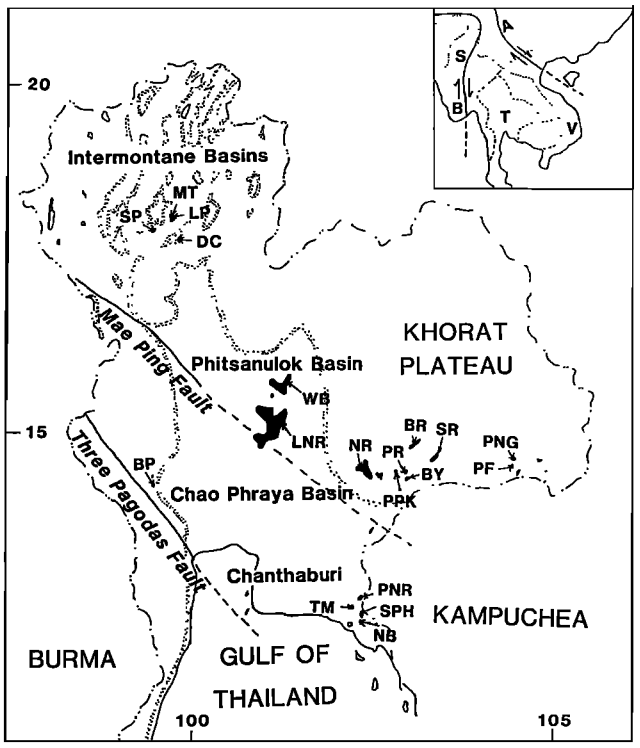


Fig. 1 Simplified map of Thailand. Exposures of late Neogene volcanics are shown as darkened areas. The letters assigned to each of the darkened regions are abbreviations given by *Jungyusuk and Sirinawin*, [1983], after *Barr and MacDonald* [1981]), for the individual locality names. These names, their abbreviations, and age (if reported) are given in Table 1. Two transcurrent faults with a NW-SE trend are shown on the map [after *Bunopas*, 1981]. Because the sense of offset on these faults is subject of controversy (see text), no sense of motion is indicated on the figure. Insert shows regional map of Indochina. Letters on map refer to areas discussed in text. These are A, Red River fault; B, Burma; S, Sagaing fault; T, Thailand; and V, Vietnam.

vena and Clark, 1986], the Phitsanulok Basin [*Nieuwland*, 1984], and the intermontane basins [*Snansieng and Maneekut*, 1985]; (2) heat flow values from both the Gulf of Thailand [*Matsubayashi and Uyeda*, 1979; *Anderson*, 1980; *Hellinger et al.*, 1981; *Matsubara et al.*, 1982; *Travena and Clark*, 1986] and the intermontane basins of western Thailand [*Barr et al.*, 1978; *Thienprasert et al.*, 1978; *Sertsrivanit*, 1984] are anomalously high with respect to both the normal geothermal gradient for continental regions and the Khorat Plateau [*Thienprasert et al.*, 1978; *Sertsrivanit*, 1984]; (3) numerous exposures of late Neogene tholeiitic basalt flows and alkalic basalt flows and related differentiates occur throughout the region [*Barr and MacDonald*, 1979, 1981; *Jungyusuk and Sirinawin*, 1983]; and (4) in central Thailand, rhyolites of late Neogene age have been found near basalt flows, suggesting a bimodal basalt and rhyolite association (R. McCabe, unpublished field notes, 1986, 1987).

The occurrences of high-angle normal faults, high heat flow, tholeiitic basalt and alkalic basalt volcanism, and related bimodel volcanism are noted in other continental regions of the world undergoing active extension. For example, similar events have been recorded in the western United States [*Christiansen and Lipman*, 1972; *Best and Brimhall*, 1974; *Eaton*, 1979] and the East African rift [*Gass*, 1970]. In these and other examples of continental rifting, volcanic activity and block faulting are closely associated in time and space. The similarity of other regions with the above observations from Thailand leads us to infer that a major phase of extension of the central basins of Thailand occurred during the Neogene.

In this paper we present Neogene paleomagnetic data from Thailand. The data suggest that the region can be partitioned into two (or more) tectonic blocks. The eastern block, which consists of the Khorat Plateau, shows Neogene paleomagnetic directions which are indistinguishable from the geocentric axial dipole field direction. The magnetic declinations from the Khorat Plateau are consistent with Neogene and Quaternary paleomagnetic directions reported from 28 sites of volcanic rocks in Vietnam [*Giang*, 1982]. The data from the Khorat Plateau and Vietnam suggest that eastern Indochina, including the Khorat Plateau, is behaving as a single tectonic block which has not rotated with respect to the geocentric dipole field. In contrast to these data from eastern Indochina, similar-aged samples collected from central and western Thailand show significant deflections in declinations with respect to the geocentric dipole field direction. The contrasting declinations suggest that Indochina is broken into a number of smaller blocks which have experienced independent rotations. We suggest that this rotation is associated with a major middle to late Neogene phase of opening of the basins of central and northwest Thailand.

Tectonic Setting

Thailand is located along the southern margin of southeast Asia. Geologic investigations [*Bunopas et al.*, 1978; *Hamilton*, 1979; *Mitchell*, 1981; *Bunopas*, 1981] suggest that Thailand is composed of two sutured terranes, the Shan-Thai Terrane and the Indochina Cratonic Terrane. These terranes are separated by a thin, north trending block of ophiolites and melanges called the Nan Suture [*Mitchell*, 1981; *Hutchison*, 1975; *Barr and MacDonald*, 1987]. The Shan-Thai Terrane is exposed in the western highlands of Thailand and the eastern highlands of Burma. The basement of this terrane consists of Precambrian granitoids, gneisses, and schists which are overlain by a complex belt of folded Paleozoic and Mesozoic rocks that crop out from northern Thailand through Malaysia. In eastern Thailand, the Indochina Cratonic Terrane is composed of relatively flat-lying Mesozoic continental sedimentary sequences of the Khorat Group [*Brown et al.*, 1951; *Bunopas*, 1981], Permian carbonate platform and deepwater facies [*Wielchowsky and Young*, 1985], and older Paleozoic rocks. The Mesozoic Khorat sedimentary rocks are believed to overlay the older Kontum Massif, which trends eastward into Loas, Kampuchea, and Vietnam. The Shan-Thai Terrane and ophiolite sequence are currently separated from the Indochina Cratonic Terrane by a central basin (Chao Phraya-Phitsanulok Basin) filled with Cenozoic sedimentary rocks. This Cenozoic basin is filled with consolidated shale, mudstones, sandstones, limestones, diatomite, coal, and unconsolidated sediments [*Brown et al.*, 1951; *Bunopas*, 1978, 1981; C. C. Wielchowsky, personal communications, 1987]. Reports on the structure of these basins show that they are relatively simple fault-bounded basins that contain flat-lying to gently folded strata [*Bunopas*, 1981].

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In and around the basins of central Thailand are numerous basalt flows. These flows are part of a large field of late Cenozoic continental basalts which cover large portions of Vietnam, Loas, Kampuchea, and Thailand. Geochemical studies and petrographic descriptions of these basalts [*Barr and MacDonald*, 1979, 1981; *Jungyusuk and Sirinawin*, 1983] show that they are tholeiitic and alkalic basalts with compositions ranging from nephelinite to tholeiite. Fission track and potassium-argon dating methods [*Barr and MacDonald*, 1981] yield ages between 0.4 and 11.3 Ma, suggesting that this entire volcanic field is of middle Miocene to Recent age.

These late Cenozoic basalts crop out as small scattered sheet flows. We observed one to five flows exposed in each of these areas. In some areas, soil horizons and well-developed chilled margins were noted between individual flows. This suggests that these flows were not erupted in rapid succession but rather were extruded intermittently. In some localities (central and southeastern Thailand), rhyolites are closely associated with these basalts. These rhyolites are assigned Paleozoic and Mesozoic ages (1:1,000,000 map of Thailand [*Thailand Department of Mineral Resources*, 1982]). However, these rhyolites contain fresh glass and are seen to rest directly upon basalts with late Neogene K-Ar ages or upon young stream gravels (R. McCabe, unpublished field notes, 1986). Furthermore, K-Ar dating has yielded a 2.1-Ma date on one of these rhyolite flows (F. Winkler, personal communications, 1987). These facts argue that at least some of these rhyolites are of late Cenozoic age.

Many faults can be delineated on aerial photographs. Two large, northwestward trending transcurrent faults, the Mae Ping fault (Papun fault of *LeDain et al.* [1984]) and the Three Pagodas fault, cut across Thailand (Figure 1). The age and sense of slip on these faults are still a subject of controversy. Offsets of various rock units across the fault show a significant amount of left-lateral separation [*Bunopas*, 1981; *LeDain et al.*, 1984; C. C. Wielchowsky, personal communications, 1987]. However, there is little direct age control on the timing of motion on these faults. *LeDain et al.* [1984] using Landsat images noted left-lateral separation of bedding planes of older strata. C. C. Wielchowsky (personal communications, 1987) noted that minor faults, with similar orientation to these major faults, cut Mesozoic-age beds, which suggests these faults were active during the Tertiary. *Bunopas* [1981] argues that these faults cut late Tertiary rocks and proposes that they may presently be active or that their activity has only recently ceased. From these observations, it appears that these faults were active during the Tertiary and may have been active in the late Tertiary. A recent focal mechanism reported by *LeDain et al.* [1984] indicates that these faults may now have a right-lateral slip sense. LeDain et al. believe this new direction of slip may be related to a late Cenozoic reactivation of these faults caused by a major change in the orientation of late Cenozoic stresses in the southeast Asian region.

PREVIOUS PALEOMAGNETIC STUDIES

Most previously reported paleomagnetic studies in Indochina are concerned with pre-Cenozoic reconstructions of the region. These studies have used Mesozoic sedimentary rocks from Thailand as their source of data [*Haile and Tarling*, 1975; *Bunopas*, 1981; *Bunopas et al.*, 1978; *Achache and Courtillot*, 1985; *Barr et al.*, 1978; *Maranate and Vella*, 1986]. Although some of these studies were based on limited data, all of the workers agree that Indochina has been near its present latitude since Jurassic time and has rotated about 24° clockwise with respect to Eurasia during the past 100 m.y. However, each author has a different opinion on when the rotation occurred. *Maranate* [1982] argues that the clockwise rotation of Indochina occurred after the Middle Cretaceous. Similarly, *Bunopas* [1981] suggests that the region underwent a progressive clockwise rotation during the Mesozoic. In contrast, *Achache et al.* [1983] and *Achache and Courtillot* [1985] argue that a large clockwise rotation occurred since the earth Tertiary as a consequence of the collision between Eurasia and India.

Very little work has been carried out on Tertiary rocks from Thailand. *Barr and MacDonald* [1979] and *Barr et al.* [1976] reported results from late Tertiary age basalts which lie unconformably over the older continental rocks of Thailand. Their studies emphasized the use of magnetostratigraphy to better constrain the age of these flows. Results from the Denchi region [*Barr and MacDonald*, 1979] are presented in Table 1. Although the data show a clear clockwise deflection from the dipole field direction, Barr and MacDonald noted that because they sampled only five flows, the anomalous declination was likely caused by secular variation of the Earth's magnetic field.

Giang [1982] reported paleomagnetic directions from late Cenozoic basalts from Vietnam (Table 2). Although these Vietnam results show a shallow mean inclination (I_{ob} = 18.9° ±4.9°) with respect to the expected dipole value (I_{ex} = 26.5°), the mean declination (D_{ob}=357.8°) is indistinguishable from the dipole declination. The shallow inclination values have also been noted in other southeast Asian regions and are attributed by *McCabe et al.* [1987] to a g_2^0 effect [*Wilson*, 1970, 1971; *Merrill and McElhinny*, 1977]. The declination data suggest that Vietnam has not rotated with respect to the dipole field direction during the past 5 m.y.

PALEOMAGNETIC TECHNIQUES AND RESULTS

Sampling and Laboratory Techniques

The results from a total of 304 independently oriented samples collected from 48 individual flows from the late Cenozoic basalts are shown in Table 1. These flows show dips of 5° or less, and therefore no bedding correction was required or applied. Samples from 29 of the flows were collected from the Khorat Plateau. Fifteen flows were sampled in central Thailand and four flows in western Thailand. The four flows from western Thailand are grouped with five flows reported by *Barr and MacDonald* [1979] from the Denchi area (Figure 1) and are listed in Table 1. The locations of the flows sampled are shown in Figure 1. Cores (25-mm diameter) were drilled at each site using a hand-held portable rock drill and were oriented with an orienting device and a Brunton compass. Five to nine (typically seven) samples were collected per flow. Sighting large topographic features with the Bruton compass and comparing the directions with map directions allowed us to make corrections for local anomalies of the Earth's magnetic field. In no case did we find that either of these directions varied by more than a few degrees. Thin section examination of the samples col-

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TABLE 1. Late Cenozoic Paleomagnetic Results From Individual Basalt Flows From Thailand

Basalt	Locality			N	D_m	I_m	α_{95}	K	VGP	
	Latitude	Longitude	Site						Latitude	Longitude
Khorat Plateau (Eastern Thailand)										
Phu Phra Angkhan (PPK)	N14°30'	E102°50'	S-6	7	187.5	-53.1	2.7	498.3	69.7	120.9
	N14°30'	E102°50'	S-7	7	183.1	-47.7	5.4	203.2	75.4	113.7
	N14°32'	E102°50'	S-20	7	192.9	-54.3	5.7	113.9	66.6	130.3
	N14°33'	E102°51'	S-22	8	180.8	-52.5	1.2	1970.0	71.4	102.8
Nakhon Ratchasima (NR)	N14°36'	E102°12'	S-8	7	354.5	28.3	7.6	64.4	84.7	18.8
	N14°43'	E102°12'	S-9	6	351.2	27.6	4.0	286.3	81.5	13.6
	N14°43'	E102°16'	S-10	6	187.5	-43.6	6.6	103.2	77.1	134.3
	N14°43'	E102°16'	S-11	7	183.7	-39.0	4.8	159.9	81.8	127.0
	N14°43'	E102°16'	S-12	5	186.4	-36.8	10.5	53.8	81.3	142.5
Khao Pha Nom Rong (PR)	N14°33'	E102°57'	S-13	5	216.1	-41.3	7.7	98.9	54.8	172.2
	N14°33'	E102°57'	S-14	7	15.4	31.4	2.4	643.3	75.0	181.4
	N14°33'	E102°57'	S-15	7	15.8	30.5	2.3	716.4	74.7	183.8
	N14°33'	E102°57'	S-16	6	16.1	36.1	13.3	26.4	73.7	170.8
	N14°31'	E102°55'	S-30	7	26.0	29.7	17.2	20.8	64.9	186.3
Khao Kradong (BR, 0.9 Ma)	N14°57'	E102°16'	S-17	7	354.8	28.8	3.6	278.9	84.2	16.7
	N14°57'	E103°07'	S-18	6	353.5	33.8	3.2	433.1	82.9	43.4
	N14°56'	E103°06'	S-19	7	356.6	27.5	2.8	453.1	86.7	6.4
	N14°24'	E102°54'	S-23	6	173.3	-3.9	7.2	88.1	69.3	337.2
Surin (SR)	N14°46'	E103°23'	S-24	7	7.6	5.3	4.2	211.7	75.5	251.1
	N14°46'	E103°23'	S-25	6	9.6	0.7	2.7	620.7	72.8	249.1
	N14°46'	E103°22'	S-26	6	191.5	1.6	6.2	119.1	70.7	246.2
	N14°46'	E103°22'	S-27	7	8.5	4.5	1.9	974.4	73.2	252.5
Prai Bat (BY)	N14°29'	E102°58'	S-28	7	205.1	-41.3	3.3	409.5	64.7	167.9
	N14°28'	E102°58'	S-29	6	187.9	-35.5	2.5	724.3	80.9	157.7
Phu Fai (PF, 3.3 Ma)	N14°38'	E104°30'	S-31	7	179.9	-2.7	3.8	253.6	76.6	284.9
	N14°38'	E104°30'	S-32	7	342.6	23.1	5.2	132.1	72.9	8.3
Phu Ngoan (PNG)	N14°43'	E104°31'	S-33	4	171.2	-11.3	15.7	35.0	77.6	329.6
	N14°43'	E104°31'	S-34	5	175.8	-21.7	3.9	379.2	84.6	334.9
	N14°43'	E104°31'	S-35	5	166.0	-17.7	3.8	408.9	75.2	353.7
Chao Praya Basin (Central Thailand)										
Sae O (SO)	N13°36'	E102°24'	S-55	6	36.5	13.1	7 3	132.7	53.5	199.6
	N13°42'	E102°13'	S-53	6	211.5	-35.8	10.0	46.0	59.2	176.2
	N13°43'	E102°30'	S-54	8	220.0	-33.3	6.1	171.2	51.4	180.5
Tha Mae (TM, 0.4 Ma)	N12°37'	E102°03'	S-39	7	17.6	18.7	8.6	144.8	72.5	200.1
	N12°36'	E102°05'	S-40	5	183.2	-32.4	4.2	332.0	84.0	152.9
Pong Nam Rong (PNR)	N12°55'	E102°23'	S-48	6	183.2	-18.4	10.1	58.4	85.3	239.8
Khlung (SPH)	N12°35'	E102°22'	S-43	7	8.0	3.7	6.7	189.8	76.6	245.4
	N12°35'	E102°15'	S-44	6	53.9	22.6	15.2	26.4	37.4	186.9
Nong Bon (NB, 1.3 Ma)	N12°35'	E102°30'	S-52	6	14.2	48.6	12.9	27.8	68.5	138.2
Lamnarai (LNR, 11.3 Ma)	N15°01'	E101°00'	J-1	8	193.9	-0.8	3.2	297.9	73.8	222.0

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TABLE 1. (continued)

Basalt	Locality			N	D_m	I_m	α_{95}	K	VGP	
	Latitude	Longitude	Site						Latitude	Longitude
Wichian	N15°45'	E100°12'	J-6	5	9.9	42.6	6.8	126.8	77.1	144.7
Buri (WB)	N15°45'	E100°12'	J-7	6	53.9	39.1	14.1	23.9	38.9	174.1
	N15°45'	E100°12'	J-8	6	197.9	-14.7	7.7	76.9	70.6	213.5
	N15°45'	E100°12'	J-9	5	193.4	-33.6	3.2	579.9	76.9	176.7
	N15°45'	E100°12'	J-10	7	191.9	-27.7	4.5	177.9	78.5	193.8
Western Thailand										
Mae Tha (MT)	N18°10'	E99°34'	S-3	7	14.5	26.2	4.2	202.7	75.4	204.8
Sop Prap (SP)	N18°06'	E99°21'	S-2	7	231.2	-10.8	7.2	79.6	38.5	196.9
	N18°06'	E99°21'	S-5	5	228.3	-21.5	21.0	14.3	42.9	190.8
Bo Phoi (BP, 3.1 Ma)	N14°10'	E99°30'	S-56	6	39.8	18.0	7.7	76.1	50.8	191.5
Western Thailand [From *Barr and MacDonald*, 1979]										
Denchai (DC, 5.6 Ma)	N17°50'	E100°04'	F1	11	199.4	-34.0	10.0	22.0	71.6	185.0
	N17°50'	E100°04'	F2	5	194.2	-25.7	14.3	29.5	75.6	206.1
	N17°50'	E100°04'	F4	1	201.0	-29.0	*	*	69.7	194.0
	N17°50'	E100°04'	F5	4	200.8	-23.8	24.0	15.7	69.2	202.6
	N17°50'	E100°04'	F7	2	206.6	-33.5	*	*	64.8	182.0

N, number of samples; D_m, mean declination; I_m, mean inclination; α_{95}, radius of 95% confidence circle; K, precision parameter; site localities are shown in Figure 1; ages assigned to various localities are K-Ar and fission track ages from *Barr and MacDonald* [1981].

*Not calculated because of limited sample number/flow.

lected shows that they are free of secondary alteration. On the basis of K-Ar and fission track age dating [*Barr and MacDonald*, 1981], the flows reported herein range in age from 11.3 to 0.4 Ma.

The remanent magnetic directions of these samples were measured using a spinner magnetometer (Minispin by Molspin Ltd.) at Texas A&M University. Only one sample was used for each independently oriented core collected in the field. Stepwise alternating field (AF) demagnetization was carried out in 5 millitesla (mT) steps from the natural remanent magnetization (NRM) to 20 mT and then in 10 mT steps from 20 to 100 mT on two to four samples from each flow using a Schonstedt GSD-1 single-axis demagnetizer. Stepwise thermal demagnetizations were carried out on a number of samples using a Schonstadt DST-1 thermal demagnetizer in steps of 50 °C from 200 ° to 550 ° and then at 580 ° and 610 °. Where warranted by stable behavior (discussed below), abbreviated demagnetization sequences were used to clean the remaining samples from each flow. Orthogonal vector plots were constructed in order to observe the

TABLE 2. Summary of Mean Neogene Paleomagnetic Directions, Indochina

	N	D_m	I_m	α_{95}	k	Pole		A_{95}	K
						Latitude	Longitude		
1.	53	13.5	27.8	5.2	15.1	79.0	187.6	4.5	19.6
2.	29	4.3	28.5	6.6	17.2	85.7	171.4	5.4	25.6
3.	24	24 4	26.2	6.9	19 2	66.4	189.5	6.2	23.5
4.	28	357.8	18.6	4.9	54.0	85.8	319.9	3.8	51.8

N, number of sites; D_m, mean declination; I_m, mean inclination; α_{95}, radius of 95% confidence circle; k, precision parameter; A_{95}, radius of 95% confidence of VGP's; and K, precision parameter for VGP's.

1, Mean direction for all of Thailand data; 2, mean direction for sites from Khorat Plateau; 3, mean direction for sites located in western and central Thailand (including sites from *Barr and MacDonald*, [1979]); and 4, Vietnam [*Giang*, 1982].

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behavior of the magnetic components of the samples during stepwise demagnetization. In each of the reported samples, a characteristic component of magnetization was considered isolated when the orthogonal projections showed the magnetic vector decaying unidirectionally toward the origin with progressive demagnetization and when less than 15% of the initial magnetization remained. Direction means of individual flows and of several flows, confidence levels (α_{95}), and precision parameters (κ) were calculated using the method of *Fisher* [1953]. The uncertainty in declinations (δD) is defined as

$$\delta D = \pm \sin^{-1} (\sin \alpha_{95}/\cos (\text{inclination}))$$

after *Wilson and Cox* [1980].

In doing paleomagnetic studies, it is important to compare directions against a reference pole. Although no Neogene reference pole has been reported for Indochina, *Achache et al.* [1983] have reported a Eurasian Mio-Pliocene pole. However, Achache et al.'s Mio-Pliocene pole is based only on data from western Europe and lacks data from Asia. We will not use Achache et al.'s late Neogene pole because of the lack of Asian data; instead we will compare our directions against the geocentric dipole field direction. Using the geocentric dipole direction as a reference pole should not introduce any errors into our discussion below because we will discuss the data in terms of the relative rotation of western and central Thailand data with respect to the data from the Khorat Plateau and Vietnam.

Magnetic Results

AF demagnetization between 10 and 40 mT effectively eliminated secondary overprints. An orthogonal vector plot from a typical sample is shown in Figure 2*a*. Because the AF demagnetization method was effective in isolating a single direction of magnetization for all but two of the flows reported herein, this method was used throughout the study. Samples from the two flows for which AF demagnetization was ineffective in removing the secondary components of magnetization were treated with thermal demagnetization. Heating at 300 °C was an effective level for eliminating the low blocking temperature, high coercivity phase(s) from all of the samples from these flows. Above 300 °C, all of the samples from these two flows showed the characteristic unidirectional decay on further levels of thermal demagnetization. Thermal demagnetization was also carried out on many of the sites that yielded stable directions to AF demagnetization. In all cases, the direction obtained by thermal demagnetization was indistinguishable from the mean direction obtained from AF demagnetization. The mean directions for the individual flows are listed in Table 1 and shown in Figure 2*b*. In the figure, reversed sites are flipped to their antipodal position.

We believe that the directions of magnetization reported in Table 1 are primary directions and do not represent secondary overprints. This statement is made on the basis of the following observations: (1) the lower coercivity magnetic components from samples from a few of these sites showed varying degrees of scatter which were removed by demagnetization, (2) thin section studies show that these rocks are free from secondary alteration, (3) both normal and reversed directions which were antipodal at the 95% confidence level, were found in different flows from the same locality, and (4)

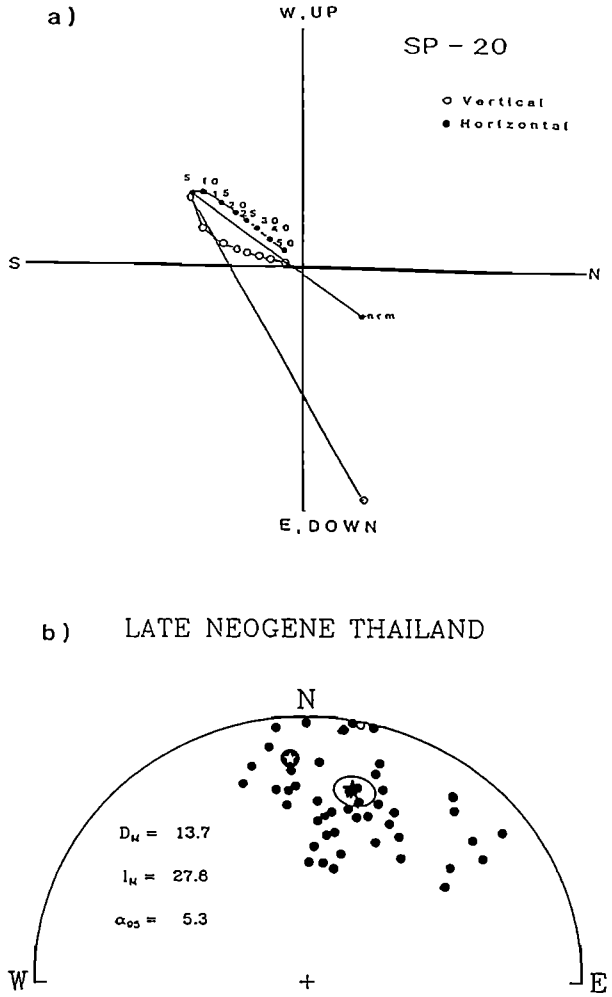


Fig 2. (*a*) Typical orthogonal vector projections of the AF demagnetization behavior of the magnetization of our samples reported in Table 1. Values of demagnetization are in millitesla. Thermal demagnetization techniques also produced similar unidirectional decay plots after heating to 300 °C (a typical vector plot for thermal is not shown). On heating to 580 °C, less than 10% of the magnetization remained on all of the samples. (*b*) Equal-area stereoplot of mean directions from 53 flows reported in Table 1. On this plot, reversed sites were flipped to their antipodal position. The overall total mean and its cone of confidence for these flows are shown by a solid star. The starred circle shows the mean direction for 28 late Neogene flows reported from Vietnam [*Giang*, 1982].

internally consistent results over a broad sampling region (> 1000 square kilometers).

When all the data presented in Table 1 are averaged by Fisher statistics, a mean direction (D_m = 13.5°, I_m = 27.8°, α_{95}=5.2°, κ=15.1) is found. Although this mean direction is distinguishable from the dipole field direction, some degree of caution should be exercised in interpreting this direction as being caused solely by tectonic rotations. Because we are working with volcanic rocks, secular variation can also be an important factor in our scatter. The degree to which our data average out secular variation can be estimated by comparing the precision parameter (κ) predicted by a model for paleosecular variation with the precision parameter calculated from our data [*Cox*, 1969]. The overall mean direction (D_m = 13.5°, I_m = 27.8°, α_{95} = 5.2°) obtained by using all the mean site directions

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($N = 53$) has a precision parameter of $\kappa = 15.1$, and at the 95% confidence level $12.6 < \kappa < 23.6$ [*Cox*, 1969]. The observed value of κ does not differ at the 95% confidence level from the expected $\kappa = 21.8$ for the overall mean paleolatitude [*McFadden*, 1980]. This preliminary analysis together with the fact that the data set reported in Table 1 consists of 53 independent flows which show mixed polarities and span over 10 m.y., all suggest that errors in our data caused by insufficient averaging of secular variation are small.

INTERPRETATION OF DATA

The mean direction from the 53 flows listed in Table 1 ($D_m = 13.5°$, $I_m = 27.8°$, α_{95}=5.2°, κ=15.1) is statistically distinguishable from the the dipole field direction. For the reasons stated above, we do not believe that this rotation is the result of insufficient time averaging of secular variation of the Earth's magnetic field. Because the observed inclination is very close to the expected dipole inclination for this region ($I_{ex} = 28.2°$ for 15 °N latitude), it appears that the deflection of the mean direction from the dipole direction is due to a clockwise rotation in declination related to late Neogene tectonics in the region.

A closer examination of the data (Table 2) shows that the mean declination for the 29 sites from the eastern Khorat Plateau ($D_m = 4.3° \pm 7.5°$; Figure 3*a*) is indistinguishable at the 95% confidence level from both the dipole field direction and the mean declination reported by *Giang* [1982] from similar-aged rocks in Vietnam ($D_m = 357.8° \pm 5.2°$). In contrast, the mean declination of similar-aged volcanics from the Chao Phraya-Phitsanulok Basin and western Thailand (Figure 3*b*) is deflected significantly clockwise ($D_m = 24.4° \pm 7.7°$) from the dipole field direction. The data suggest that Indochina is divided into two or more crustal blocks which have different Neogene translational histories.

Figure 4 shows late Neogene poles from Eurasia [*Achache et al.*, 1983], Vietnam [*Giang*, 1982], the Khorat Plateau, and western and central Thailand. The data from the Khorat Plateau are indistinguishable at the 95% confidence level from the geocentric dipole field, the late Neogene pole from Vietnam, and from Achache et al.'s Mio-Pliocene pole. In constrast to the late Neogene poles from the Khorat and Vietnam, the data from central and western Thailand are distinguishable at the 95% confidence level from both these poles and from the Mio-Pliocene pole of *Achache et al.* [1983].

DISCUSSION

Tertiary Tectonics of Indochina

As discussed above, paleomagnetic vectors from Thailand are rotated clockwise from the expected geocentric dipole direction for this region. There are at least four possible explanations that have been proposed which could result in a Tertiary clockwise rotation of this region. These models are (Figure 5) as follows.

1. Extrusion tectonics model (Figure 5*a*). The Cenozoic tectonic development of the southeast Asian margin has been related to the Eocene collision between the Indian Craton and the southern margin of the Eurasian continent [*Molnar and Tapponnier*, 1975; *Tapponnier and Molnar*, 1976, 1977; *Curray et al.*, 1979; *Hamilton*, 1979; *Mitchell*, 1981; *Tapponnier et al.*, 1982]. On the basis of plasticine experiments, *Tapponnier et al.* [1982] produced a tectonic model (the extrusion model) which suggests that the large strike-slip faults of eastern Asia are large shear zones which allow southeast Asia to be extruded southeastward, in a sense similar to the motion of Indochina proposed earlier by *Rodolfo* [1969]. Using this model, Tapponnier et al. propose that the Tertiary tectonics of eastern Asia is dominated by two major phases of continental extrusion. In the first stage (50–20 Ma), Indochina was extruded to the southeast and rotated clockwise 25°. The second phase of extrusion (which began around 20 Ma to the present) resulted in a clockwise rotation of an additional 15°.

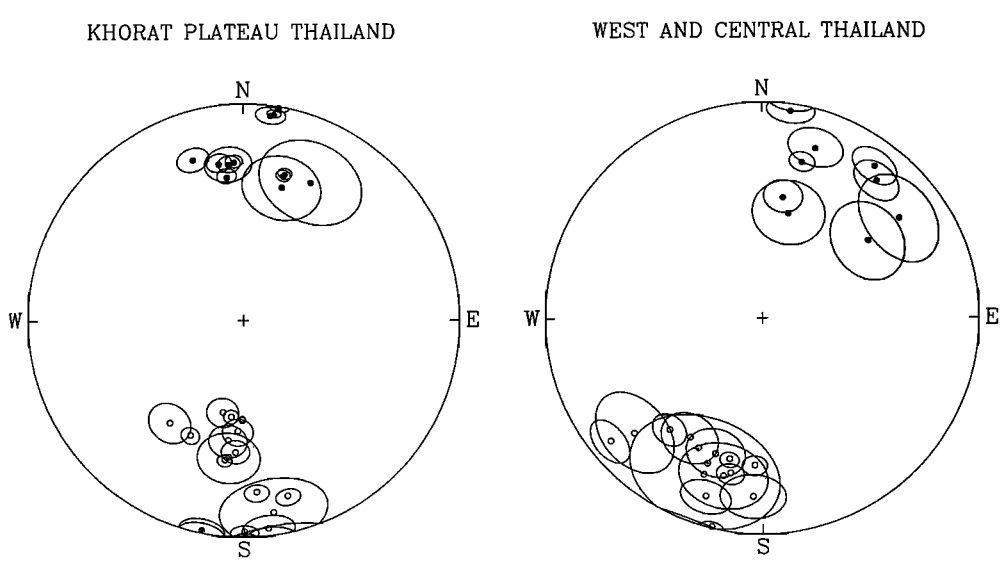


Fig. 3. Equal-area stereoplots of mean direction from individual flows and their cones of 95% confidence are shown for data from (*a*) the Khorat Plateau and (*b*) western and central Thailand. The mean direction for the the Khorat Plateau is indistinguishable from the geocentric dipole direction. In contrast, the overall mean direction of flows from western and central Thailand is significantly deflected clockwise from the geocentric dipole field. This deflection suggests that this region has rotated westward with respect to the Khorat during the late Neogene.

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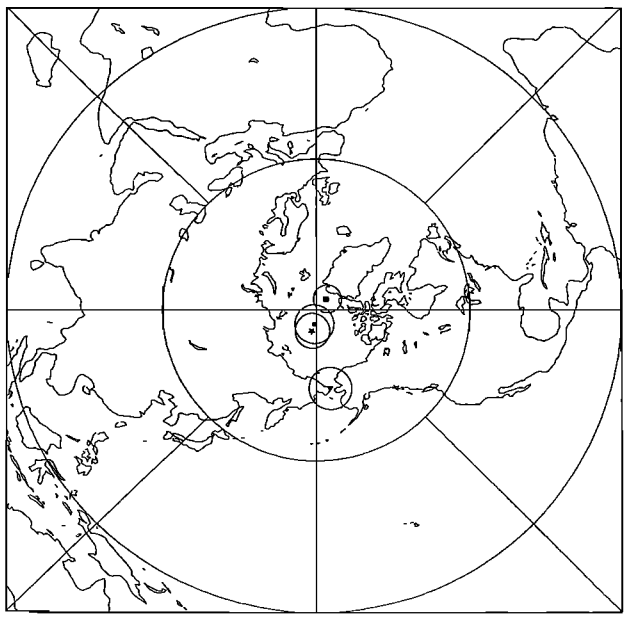


Fig. 4. Late Neogene poles from the Khorat Plateau (circle), western and central Thailand (triangle), and Vietnam (square) and their α 95 are shown. Location of the Mio-Pliocene Eurasian pole of *Achache et al.* [1983] is shown by a star. This pole was calculated with data from western Europe Because of the lack of Asian data in calculating the Eurasian pole, the geocentric dipole field was used as a reference pole for this study. The data from the Khorat Plateau are indistinguishable at the 95% confidence level from the Eurasian pole, the late Neogene Vietnam pole, and from the geocentric dipole field direction. In contrast, the late Neogene pole from western and central Thailand is clearly distinguishable from these three other poles.

2. Collision-related bending model (Figure 5*b*). Another possible explanation for the observed clockwise deflections is that they are related to the collision of India with the Asian Arc, resulting in a bending of the upper plate. This phenomenon is observed in many island arcs which show pronounced bends in regions of the arc near bathymetric features on the downgoing plate [*Carey*, 1958; *Vogt*, 1973; *Vogt et al.*, 1976]. Paleomagnetic directions near these island-arc collision zones show differential rotations which are of the same age as the collision [*McCabe et al.*, 1983; *McCabe and Uyeda*, 1983; *McCabe*, 1984], suggesting that the bending is related to the collision. The collision-related volcanic arc-bending model predicts that Indochina, which is on the upper plate and east of the Indian Craton (which is the bathymetric feature), should rotate clockwise as a result of the collision.

3. Strike-slip rotation model (Figure 5*c*). Horizontal rotations in paleomagnetic vectors may also be caused by strike-slip faults by either simple shear rotations [*Beck*, 1976, 1980] or by the rotations of small crustal blocks that are between larger parallel fault sets in the region [*Freund*, 1974; *Luyendyk et al.*, 1980]. These shear-related rotations have been inferred by numerous workers to try and explain discordant sets of paleomagnetic directions [e.g., *Cox*, 1981; *Terres and Luyendyk*, 1985; *Ron et al.*, 1984; *Fagin and Gose*, 1983; *Wells and Coe*, 1985; *Hudson and Geissman*, 1987; *Brown and Golombek*, 1985, 1986; *Golombek and Brown*, 1988]. Two strike-slip faults with a northwest trend have been mapped in Thailand; the Mae Ping fault (Papun fault of *LeDain et al.*, [1984]) and the Three Pagodas fault. A focal mechanism from along the Mae Ping fault [*LeDain et al.*, 1984] suggests that these faults are moving with right-lateral offset. Therefore the rotations we observe could be related to this dextral shear.

4. Extension model (Figure 5*d*). As pointed out by *Simpson and Cox* [1977], *Magill and Cox* [1981], *Cox* [1981], *Magill et al.* [1981], *Frei et al.* [1984], *Frei* [1986], and *Gromme et al.* [1986], paleomagnetically detectable rotations may be caused by movement of crustal blocks loosened by extension. In the case of Thailand, E-W Neogene extension of the central basins of Thailand could be responsible for the Neogene rotations we observed. It is worth noting that between the time of the first submission of this manuscript and the submission of the revised version, a noteworthy paper was published [*Wells and Heller*, 1988] which discusses models 3 and 4 above in more detail. *Wells and Heller* [1988, and the numerous references listed therein] use a combination of dextral shear and extension in the Basin and Range Province to explain the discordant Cenozoic paleomagnetic vectors reported from the northwestern United States.

The anomalous declinations we are observing have the correct clockwise rotation predicted by the extrusion model and the island arc bending model (explanations 1 and 2 above). However, the rotations we observed occur during the late Miocene and the Pliocene and are too young to be consistent with either of these models. Furthermore, these two models infer that the entire Indochina region behaves as a single rigid block. If this is the case, the entire region should show similar magnetic directions for the same-aged rocks. This is not consistent with our data, which show that the western portion of Indochina is rotated nearly 25° with respect to eastern Indochina.

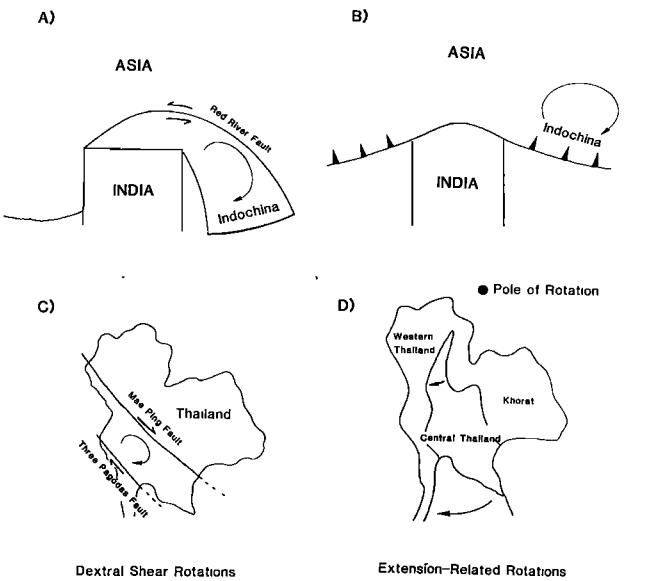


Fig. 5. Possible models to explain the observed clockwise rotations of paleomagnetic vectors in Thailand. (*a*) Extrusion model of *Tapponnier et al.* [1982], (*b*) collision-related bending model [*Carey*, 1958; *Vogt*, 1973; *McCabe*, 1984], (*c*) dextral shear rotation model [*Beck*, 1976; *Freund*, 1974; *Luyendyk et al.*, 1980], (*d*) extension model [*Simpson and Cox*, 1978; *Cox*, 1981; *Magill and Cox*, 1981; *Frei et al.*, 1984; *Gromme et al.*, 1986; *Wells and Heller*, 1988].

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Although we presently do not have enough data to rule out the possibility that the rotations we observed are related to strike-slip faulting (strike-slip rotation model), we do not favor this choice for the following reasons. Two northwest trending faults (the Mae Ping fault and the Three Pagodas fault) occur near our sampling sites. If the rotations we are observing are related to dextral motion on these strike-slip faults, we may expect rocks, which are part of the tectonic block situated between these faults, to have a different rotational history than rocks from other structural blocks. Although we still do not have enough data to properly distinguish between the rotations of the different blocks, our limited data do not appear to indicate different rotational histories for the different blocks. For example, sites tens of kilometers to the north of the Mae Ping fault (localities MT, SP, and DC in Figure 1) show rotations similar to sites located near the Mae Ping fault (SR, LNR, WB), as well as the site near the Three Pagodas fault (BP). Furthermore, because the faults show current seismic activity [*LeDain et al.*, 1984], we should expect rotations in rocks younger than 1 Ma, which we do not see. Last, *Bunopas* [1981] suggests that these faults were moving with left-lateral motion during the late Tertiary. In the future, we plan to collect more data from the southeastern section of Thailand (near Chanthaburi; Figure 1). These data should help us better understand if the different tectonic blocks in the region are undergoing independent rotations.

The geographic distribution of the rotated sites suggests that the rotations are related to the extension associated with the opening of north-south trending basins in central Thailand. This conclusion is supported by the fact that most of the flows located in both the Chao Phraya-Phitsanulok Basin (LNR, SR, WB, SPH, NB) and the mountainous western portion of Thailand (DC, MT, SP, BP) show significant clockwise rotations with respect to the dipole field. This evidence, together with the observation that the Khorat Plateau and Vietnam behaved as a single block, suggests that the areas of Indochina that are west and south of the Khorat Plateau have rotated clockwise nearly 25° with respect to the eastern portion of Indochina during the late Neogene.

Model for Development of Late Neogene Basins

We present below a speculative model for the origin of the north-south trending Tertiary basins of central Thailand. This model represents our current working model; however, we should note that it is limited by the paucity of age constraints on the late Cenozoic volcanics and the lack of sites older than 11 Ma. Although it is possible that the basins of this region underwent an earlier Tertiary rifting event, our data do not allow us to speculate on either the cause or the nature of such a pre-late Miocene extension. In the future, we plan to obtain more age data from these volcanics and to start sampling pre-middle Miocene rocks. It is hoped that these new data will either strengthen or modify the speculative model presented below.

Late Miocene and Pliocene. Crustal extension occurred in an east-west direction during the late Miocene and Pliocene. During this period, a series of north-trending extensional basins developed in which thick sequences of clastic rocks were deposited. Although the cause for this extension is unknown, we suggest that the extension was similar to extension in the Basin and Range Province of the western United States. Furthermore, we suggest that this extension is related to subduction of the Indian plate under the Andaman Islands and western Burma. Until we have better paleomagnetic data on the early and middle Miocene rocks from the region, we are unable to determine when the translation began.

An examination of Figure 1 shows that the basins become progressively wider to the south. This southward widening suggests that the initiation of extension might have begun in the south and propagated northward. However, the available sedimentological data are too limited to allow one to confirm that the deeper portions of the southern basins are older than the deepest portions of the basins to the north. On the basis of the existing data, it is just as likely that the difference in widths of the basins is related to the fact that the common pole of opening for this entire basin system was north of Thailand in central China. Such a pole would imply that rifting began at the same time in both the northern and the southern portions of western Indochina and the variation in width of the basins is proportional to the distance from the pole. In other words, locations near the pole of opening would open at a much slower rate (and therefore be narrower) than locations far from the pole. As mentioned above, the current sense of late Neogene motion on the northwest-southeast trending transcurrent faults is not known and is the subject of controversy. It would be interesting to speculate that these faults are somehow related to development of the basins. However, until more data are gathered on the sense of motion and the amounts of offsets on these faults, relating the motion on these faults to basin formation is not possible.

Associated with crustal extension are the high geothermal gradients in these basins. Geothermal gradients of 4.0°–5.8°C/100 m have been reported in the Gulf of Thailand [*Matsubayashi and Uyeda*, 1979; *Hellinger et al.*, 1981], and even higher gradients have been reported in the intermontane basins in the north [*Thienprasert et al.*, 1978; *Barr et al.*, 1978; *Sertsrivanit*, 1984]. These values are much higher than the normal geothermal gradient for sedimentary basins (2.6°C/100 m; *Travena and Clark* [1986]) and much higher than heat flow measurements reported from the Khorat Plateau [*Thienprasert et al.*, 1978; *Sertsrivanit*, 1984]. Also noted in this region are numerous fields of late Neogene basalts. Although the age of these basalts is poorly constrained, the fact that these rocks record a portion of the rotational history of the basin and have K-Ar dates [*Barr and MacDonald*, 1981] similar in age to the major basin bounding faults [*Hamilton*, 1979; *Nieuwland*, 1984; *Travena and Clark*, 1986] suggests that the basalts were erupted contemporaneously with basin formation. Our paleomagnetic data show that the basalts along the western margin (D_m=28.8°±10.7°) are deflected a few degrees more from the dipole field direction than basalts in the basin (D_m = 21.6°±11.1°). However, these data are not statistically distinguishable from each other, and hence we are unable to discern if there is an actual declination difference between the western and central portions of the basin. In the future, we plan to collect more data from these regions.

Quaternary. The limited number of data from sites whose ages of Pleistocene and Holocene arefrom all of Thailand are not significantly rotated with respect to the dipole field. This suggests that the rotations associated with basin for-

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mation were over by the Quaternary. Unlike the basalts, the occurrence of young rhyolites in Thailand is restricted to the Chao Phraya-Phitsanulok Basin (near the Lamnarai region, Figure 1) and the northeastern portion of the Gulf of Thailand (small islands located south of Chantiburi, Figure 1). We suggest that these rhyolites were erupted in the latest Pliocene to Pliestocene as part of a late-stage bimodal phase of volcanism. This bimodal phase is perhaps related to crustal contamination of the magma source caused by extension-related crustal thinning under the central basin.

CONCLUSIONS

1. Paleomagnetic vectors collected from late Neogene volcanic rocks of Thailand are deflected significantly clockwise from the dipole field direction.

2. The late Neogene paleomagnetic declinations from the eastern Khorat Plateau are consistent with reported declination data from Vietnam [*Giang*, 1982] and are indistinguishable from the dipole field direction.

3. In contrast, paleomagnetic directions from the central Chao Phraya Basin and from the mountains west of the basin are distinguishable from the dipole field direction. These regions are rotated nearly 25° clockwise with respect to the Khorat Plateau. Since the observed inclination values from this region agree closely with the expected inclination for this latitude, the deflections are caused by anomalies in the declination value. This suggests that the cause of rotation is tectonic in origin.

4. Although the rhyolites found in the basin are believed to be Paleozoic and Mesozoic in age [*Thailand Department of Mineral Resources*, 1982], we have observed that these rhyolites sit directly upon stream gravels and directly overlie late Neogene basalts. In addition, the late Neogene basalts from this region show chemical affinities [*Barr and MacDonald*, 1981] similar to volcanic rocks from other regions of the world undergoing extensional tectonics [*Christiansen and Lipman*, 1972]. These observations suggest that some of the rhyolites are very young and likely form part of a bimodal volcanic suite with the alkalic basalts. We suggest that the eruption of these rhyolites is associated with the waning phase of basin extension.

5. Last, a major phase of east-west extension occurred in this region during the late Neogene. This statement is based on the following geological and geophysical observations: (1) differential late Neogene tectonic rotations between the Khorat Plateau and west-central Thailand, (2) the anomalously high heat flow for parts of the region, (3) late Neogene alkalic basaltic fields with associated rhyolites, and (4) late Neogene north trending normal faults within and along the margins of the basins.

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